

Yonghao Zhao

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Education

University of Colorado Anschutz Medical Campus

Aurora, CO, USA

Ph.D. in Computational Bioscience

Aug 2026 – Present

University of Leeds

Leeds, UK

B.Sc. in Computer Science

Sep 2022 – Jun 2026

Southwest Jiaotong University (SWJTU)

Chengdu, China

B.Eng. in Computer Science

Sep 2022 – Jun 2026

Honors and Awards

- **Hirs Merit Award** (\$5,000 academic enrichment scholarship), University of Colorado School of Medicine, 2026.
- **First-Class Honours**, University of Leeds, 2026.
- **Best Bachelor's Thesis Award**, Southwest Jiaotong University, 2026.

Research Interests

Biomedical machine learning, survival and trajectory analysis, multimodal representation learning, biological foundation models, and large-language-model workflows for clinical research.

Publications

1. **Zhao, Y.**, Zheng, T., Yu, J., & Li, H.
COUPLE: Contrastive Fused Gromov–Wasserstein Optimal Transport for Scalable Cross-Modal Spatial-Omics Alignment.
Manuscript under review, 2026.
2. **Zhao, Y.**, You, Z., Shen, Y., Chu, J., Gong, X., Li, T., Wang, Z., Xu, C., Luo, Z., & He, Y.
DeSCENT: Deconvolutional Single-Cell RNA-seq Enhances Transcriptome-based Cancer Survival Analysis.
[bioRxiv preprint](#), 2026. [Under revision]
3. **Zhao, Y.**, Zhu, Z., Yang, S., & Li, W.
YeastSAM: A Deep Learning Model for Accurate Segmentation of Budding Yeast Cells.
Molecular Biology of the Cell, 2026. [Featured by [ASCB News](#)]
4. **Zhao, Y.**, Li, C., Shu, C., Wu, Q., Li, H., Xu, C., Li, T., Wang, Z., Luo, Z., & He, Y.
Tackling small sample survival analysis via transfer learning: A study of colorectal cancer prognosis.
Artificial Intelligence in Medicine, 2026.

Research Experience

Cross-Modal Spatial-Omics Alignment

Hangzhou, China

Research Assistant, Westlake University; advised by [Haoyang Li](#)

Apr 2026 – Present

- Developed COUPLE to align large, heterogeneous biomedical datasets while learning shared representations and explicit cell-level correspondences.
- Designed a candidate-sparse transport formulation with linear memory scaling, enabling whole-slide alignment of approximately 4×10^5 and 2.7×10^5 cells on a single 32 GB GPU; evaluated the method across five real modality pairs and a controlled perturbation benchmark.
- Outperformed every evaluated baseline on all five real cross-modal pairs and at every tested perturbation level; COUPLE was the only evaluated method to provide both a shared cross-modal representation and explicit cell-level correspondence at whole-slide scale.

Single-Cell Representation Learning and Multimodal Transcriptomics

Project Lead, Southwest Jiaotong University (SWJTU)

Chengdu, China

Aug 2024 – Jan 2026

- Developed [DeSCENT](#), a multimodal framework integrating bulk RNA-seq with deconvolution-guided, diffusion-generated single-cell RNA-seq for cancer prognosis, with a focus on modality alignment and biologically informed representation learning.
- Built generative modeling and evaluation pipelines for reconstructed single-cell data, including preprocessing, normalization, and distribution-level assessment of transcriptomic representations.

Cancer Survival Modeling under Low-Data Regime

Project Lead, Southwest Jiaotong University (SWJTU)

Chengdu, China

Dec 2023 – Jul 2024

- Studied time-to-event cancer outcomes across low-data and multimodal settings, spanning statistical survival models, machine learning approaches, and deep survival frameworks.
- Proposed transfer learning strategies for risk prediction under limited sample sizes and extended this line toward multimodal prognosis using transcriptomic representations from bulk and generated single-cell data.

Vision Foundation Model Adaptation for Biological Image Analysis

Project Lead, Brown University

Providence, USA

Feb 2025 – Sep 2025

- Led [YeastSAM](#), a microscopy image analysis framework based on fine-tuning Segment Anything for budding yeast cell segmentation under limited annotation and compute constraints.
- Achieved more than threefold higher accuracy in segmenting dividing cells than existing methods.
- Developed downstream modules for division-event detection, mother–bud pairing, and quantitative cellular analysis, building end-to-end experience in biological image modeling workflows.

Project Experience

[ClawBio](#): Agentic Skill System for Computational Biology Workflows

Core Contributor

Remote / China Region

Feb 2026 – Apr 2026

- Translated computational biology best practices into reusable agent skills, including scRNA-seq processing pipelines, single-cell embedding construction, and biological data visualization workflows.
- Developed and maintained skills connecting agent workflows to upstream systems and data sources, including Illumina-related inputs and BigQuery-backed access, while supporting development workflows and CI for reproducible bioinformatics automation.

Technical Skills

Languages: Python, R, C/C++

Machine Learning: PyTorch, PyTorch Geometric, scikit-learn, generative modeling, multimodal learning, survival analysis, retrieval-augmented generation

Computational Biology: Scanpy, AnnData, lifelines, single-cell RNA-seq, spatial omics, transcriptomic preprocessing

Scientific Computing: NumPy, pandas, CUDA, Linux, SLURM

Tools: Microsoft Outlook, Excel, Word, Git, LaTeX, GraphPad Prism, Adobe Illustrator